1 **(a)** State Coulomb's law.

....

....

.... [2]

(b) In a model of an atom, an electron is in a circular orbit around a proton, as shown in Figure 1.1.

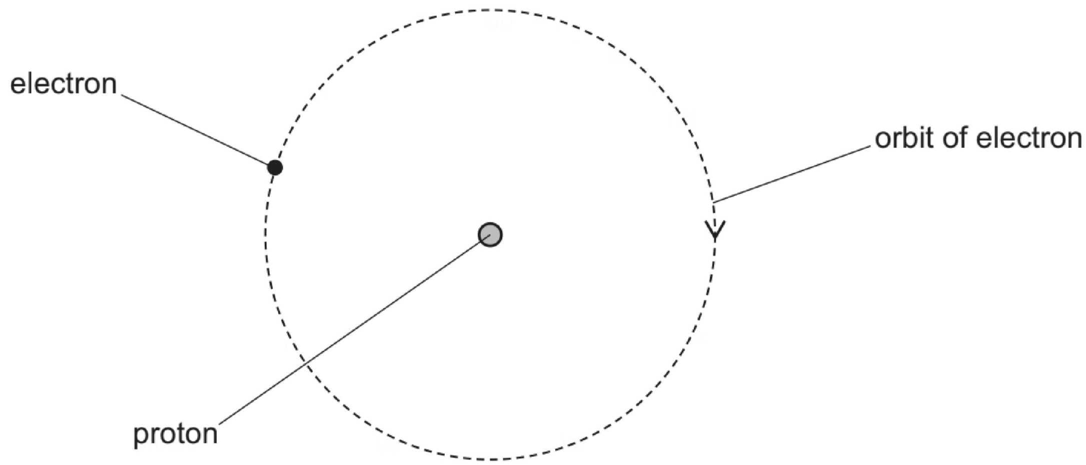

Figure 1.1

The electric potential energy of the electron is -1.09×10^{-18} J.

(i) Show that the radius of the orbit is 2.11×10^{-10} m.

[1]

(ii) Use the information in **1(b)(i)** to determine the magnitude of the electric force between the electron and the proton.

force = ... N [1]

(iii) Calculate the centripetal acceleration of the electron.

acceleration = ... $m s^{-2}$ [1]

(iv) Use your answer in **1(b)(iii)** to determine the period of the orbit of the electron.

period = ... s [2]

(c) The electron in **1(b)** emits a photon so that the electric potential energy of the electron decreases. The electron remains in a circular orbit around the proton.

By placing **one** tick (✓) in each row, complete Table 1.1 to show how the quantities indicated for the electron change when the potential energy decreases.

Table 1.1

	decreases	**stays the same**	**increases**
radius of orbit			
centripetal acceleration			
angular speed			

[3]

[Total: 10]